

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2311

Speed 138 kV
February 20, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2311. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Speed 138 kV substation in the Kentucky area. The request was received on May 09, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 458 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$154,586,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Siler Bay Power / Valliant 138.0 kV	Silver Bay Power / Damon 138.0 kV	247.0	237	26.7
56Th St / Oates Reit 138.0 kV	Speed Substation / Silver Bay Power 138.0 kV	184.9	1,170	99.9
Narod / Brockport 138.0 kV	Silver Bay Power / Speed 138.0 kV line	182.2	1,222	99.9
56Th St / Santa Monica 138.0 kV	Silver Bay Power / Speed 138.0 kV	181.9	1,210	99.9
Hambrick / Santa Monica 138.0 kV	Speed Substation / Silver Bay Power 138.0 kV	180.2	1,233	99.9
Valliant / Greenville Ave 138.0 kV	Speed / Gaga 138.0 kV	180.1	251	20.0
Narod / Barnham 138.0 kV	Silver Bay Power / Speed 138.0 kV L	179.3	1,262	99.9
Barnham / Gaga 138.0 kV	Silve Bay Power / Speed 138.0 kV	177.8	1,285	99.9
Silver Bay Power / Damon 138.0 kV	Speed / Gaga 138.0 kV L	176.5	446	33.6
Speed / Silver Bay Power Substation 138.0 kV	Gaga Substation / Speed 138.0 kV	176.4	1,307	99.9
Speed / Gaga Substation 138.0 kV L	Speed / Silver Bay Power 138.0 kV	176.4	1,307	99.9
Carraizo 1201 Substation / May Street 138.0 kV	Silver Bay Power / Speed Substation 138.0 kV L	165.9	292	27.0
Carraizo 1201 / May Street 138.0 kV Ckt 2	Silver Bay Power / Speed 138.0 kV	165.9	292	27.0
Silver Bay Power / Las Cruces 138.0 kV	Pine Bluff Watson Chapel / Carraizo 1201 345.0 kV	156.8	220	14.7

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Wagn Mound / Universiity 138.0 kV L	Pine Bluff Watson Chapel / Carraizo 1201 Substation 345.0 kV line	154.6	272	16.1
May St / Carraizo 1201 138.0 kV Line 3	Silver Bay Power / Speed 138.0 kV Line	149.6	324	27.0
Silver Bay Power / Hotel Road 138.0 kV	Speed / Gaga 138.0 kV	139.7	534	31.8
Ussc / Las Cruces 138.0 kV	Pine Bluff Watson Chapel / Carraizo 1201 345.0 kV	139.2	235	14.7
Alibates / Juliette 138.0 kV Line	Pine Bluff Watson Chapel / Carraizo 1201 345.0 kV	137.0	515	27.0
Davie / Garden Villas 138.0 kV	Desoto Pleasant Run / Union 138.0 kV line	135.6	180	5.5
Guadalupe / Mistersky 138.0 kV line	Wissota / Berry Street 138.0 kV	128.4	305	7.7
Guadalupe / Mortan 138.0 kV	Wissota / Berry St 138.0 kV	127.2	324	7.7
Mainwaring / Mortan 138.0 kV	Wissota / Berry Street 138.0 kV Line	127.0	327	7.7
Ussc / N Dracut 138.0 kV	Carraizo 1201 Substation / Pine Bluff Watson Chapel 345.0 kV	126.4	245	14.7
Pine Bluff Watson Chapel Substation / Carrizo 1201 345.0 kV	Silver Bay Power / Speed 138.0 kV Line	123.8	1,468	62.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Siler Bay Power / Valliant 138.0 kV	Rebuild line with higher-capacity conductor	\$8,628,000	67% (\$5,763,000)
Oates Reit / 56Th Street Substation 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,715,000	68% (\$1,161,000)
Brockport / Narod 138.0 kV	Rebuild line with higher-capacity conductor	\$2,310,000	68% (\$1,564,000)
56Th St / Santa Monica 138.0 kV	Rebuild line with higher-capacity conductor	\$2,949,000	68% (\$1,996,000)
Hambrick / Santa Monica 138.0 kV	Rebuild line with higher-capacity conductor	\$4,406,000	68% (\$2,983,000)
Greenville Ave / Valliat Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$4,979,000	66% (\$3,275,000)
Narod / Barnham 138.0 kV	Rebuild line with higher-capacity conductor	\$19,813,000	68% (\$13,414,000)
Barnham / Gaga 138.0 kV	Rebuild line with higher-capacity conductor	\$2,200,000	68% (\$1,489,000)
Silver Bay Power / Damon 138.0 kV	Rebuild line with higher-capacity conductor	\$3,950,000	68% (\$2,675,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Speed / Silver Bay Power 138.0 kV	Rebuild line with higher-capacity conductor	\$19,571,000	68% (\$13,250,000)
Gaga / Speed 138.0 kV	Rebuild line with higher-capacity conductor	\$5,973,000	68% (\$4,044,000)
May St / Carraizo 1201 138.0 kV Ckt 1	Rebuild line with higher-capacity conductor	\$3,861,000	62% (\$2,387,000)
May Street / Carraizo 1201 138.0 kV Circuit 2	Rebuild line with higher-capacity conductor	\$4,457,000	62% (\$2,755,000)
Silver Bay Power / Las Cruces 138.0 kV	Rebuild line with higher-capacity conductor	\$14,974,000	64% (\$9,534,000)
Wagon Mound / University 138.0 kV	Rebuild line with higher-capacity conductor	\$13,722,000	56% (\$7,669,000)
May St / Carraizo 1201 138.0 kV Line 3	Rebuild line with higher-capacity conductor	\$3,013,000	62% (\$1,862,000)
Silver Bay Power / Hotel Road 138.0 kV	Rebuild line with higher-capacity conductor	\$18,884,000	63% (\$11,957,000)
Ussc / Las Cruces Substation 138.0 kV line	Rebuild line with higher-capacity conductor	\$5,910,000	64% (\$3,763,000)
Juliette / Alibates Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$9,344,000	58% (\$5,457,000)
Garden Villas / Davie 138.0 kV	Rebuild line with higher-capacity conductor	\$13,240,000	40% (\$5,331,000)
Guadalupe / Mistersky 138.0 kV line	Rebuild line with higher-capacity conductor	\$19,195,000	32% (\$6,217,000)
Guadalupe / Mortan Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$16,262,000	32% (\$5,267,000)
Mortan / Mainwaring 138.0 kV	Rebuild line with higher-capacity conductor	\$10,044,000	32% (\$3,253,000)
Ussc / North Dracut 138.0 kV	Rebuild line with higher-capacity conductor	\$5,580,000	64% (\$3,553,000)
Pine Bluff Watson Chapel Substation / Carrizo 1201 345.0 kV	Rebuild line with higher-capacity conductor	\$59,139,000	57% (\$33,967,000)
	Total allocated to project		\$154,586,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.